**Seminar Report**

# Machine Learning - Assignment 03

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## 1 Intuition on SVD

### 1.1 Guessing the SVD

| Label | Rank | Left Singular Vectors | Singular Values | Right Singular Vectors |
|---|---|---|---|---|
| $M_1$ | 1 | $u_1 = \frac{1}{\sqrt{3}}(1,1,1,0,0)^T$ | $\sigma_1 = \frac{1}{\sqrt{3}}$ | $v_1 = \frac{1}{\sqrt{3}}(1,1,1,0,0)^T$ |
| $M_2$ | 1 | $u_1 = \frac{1}{\sqrt{3}}(1,1,1,0,0)^T$ | $\sigma_1 = 3\sqrt{3}$ | $v_1 = \frac{1}{3}(0,2,1,2,0)^T$ |
| $M_3$ | 1 | $u_1 = \frac{1}{2}(0,1,1,1,1)^T$ | $\sigma_1 = 2\sqrt{3}$ | $v_1 = \frac{1}{\sqrt{3}}(0,1,1,1)^T$ |
| $M_4$ | 2 | $u_1 = \frac{1}{\sqrt{3}}(1,1,1,0,0)^T$, | $\sigma_1 = 3$, | $v_1 = \frac{1}{\sqrt{3}}(1,1,1,0,0)^T$, |
| | | $u_2 = \frac{1}{\sqrt{2}}(0,0,0,1,1)^T$ | $\sigma_2 = 2$ | $v_2 = \frac{1}{\sqrt{2}}(0,0,0,1,1)^T$ |
| $M_5$ | 3 | failed | failed | failed |
| $M_6$ | 2 | failed | failed | failed |

### 1.2 Comparison with correct SVD

For the first four matrices the expectation of the SVD was identical with the commputed SVD (apart from a few carless mistakes, such as normalization errors). For the inutition, it helped to think about the right singular vectors as parts and the left singular vectors as weightening how much of the parts is contained in each row of the matrix. The singular values in the end determine the weight of the parts needed to reconstruct the original matrix. For $M_5$ and $M_6$, this was much trickier due to the higher rank, and because in the end each row of the matrices consist of multiple weighted parts (compared to $M_4$ where the rank is 2 as well, but each row of the original matrix can be computed by using only 1 weighted part).

### 1.3 Rank 1 Approximations

For the rank 1 matrices, the rank 1 approximation will be identical as the original matrix. For $M_4$ (see 7), the rank 1 approximation contains only the upper left 3x3 submatrix and sets all values in the lower right submatrix to 0. This is inutitive, as this is the result when replacing the second (smaller) singular value with 0.
For $M_5$ (8) and $M_6$ (9), the rank 1 approximation is much more complex to interpret, again because rows can only be computed using multiple weighted parts. Therefore, using only one of the weighted parts doesn't result in an interpretable pattern.

### 1.4 Non-singular Entries of $M_6$

Matrix 6 has rank 2 and hence 3 non-zero singular values. According to numpy, all singular values are non-zero, however. This originates from floating point arithmetics and rounding errors in numpy. $\sigma_3$ and $\sigma_4 \approx 10^{-17}$, and $\sigma_5 \approx 10^{-50}$, so they all can be assumed to be zero.

## 2 The SVD on Weather Data

### 2.1 Data Normalization

The climate matrix was normalized to z-scores, ensuring equal contribution from all features to the SVD analysis. This step is essential to account for differing units of measurement (e.g., degrees Celsius for temperature versus millimeters for precipitation) and to prevent attributes with larger scales from dominating the decomposition.

The assumption of equal importance for all attributes is reasonable, as minimum, maximum, and average temperatures, along with precipitation for each month, collectively capture seasonal variations, extremes, and overall climate dynamics necessary for comprehensive climate pattern analysis.

Regarding the normal distribution assumption, real-world climate data is not necessarily normally distributed due to natural variations and seasonal effects. Histogram analysis reveals some skewness, particularly in precipitation features. However, the distributions are approximately normal, allowing us to reasonably accept the assumption for normalization purposes.

### 2.2 SVD and Rank of Normalized Data

SVD was computed using NumPy's `svd` function, yielding the decomposition $X = U\Sigma V^T$. Fragments of matrices $\Sigma$ and $V^T$ are presented in the figures. The rank of the normalized data is **48**, revealing a lack of linear dependency between the features. The singular values decay rapidly from $\sigma_1 = 290$ to $\sigma_{48} = 7.93$, indicating that the data structure is dominated by a few principal components while maintaining full rank.

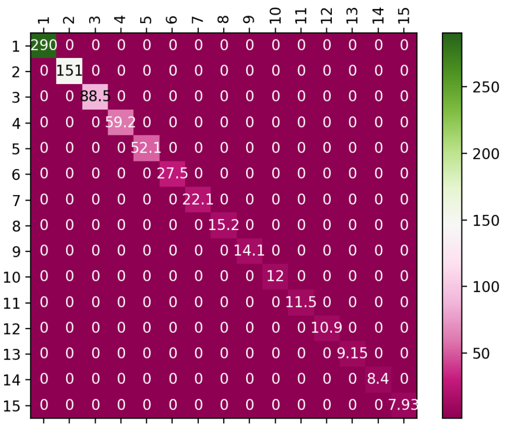


Figure 1: The first 15 singular values in diagonal matrix $\Sigma$

### 2.3 Interpretation of the First 5 Left Singular Vectors

The first 5 columns of the $U$ matrix (left singular vectors) are plotted with longitude and latitude as coordinates and entries as color values. Values of the corresponding rows of $V^T$ are taken into account for interpretation.

**First column of U:** Highlights northern regions with cooler yearly temperatures, drier winters, and wetter summers (positive values) and southern regions with warmer temperatures, wetter winters, and drier summers (negative values). $V^T$ weights suggest this represents a tendency toward low temperatures (around $-0.16$ for all temperature features) and rainy summers (0.06-0.11 for precipitation in June-September). This captures the *primary North-South temperature gradient*.

**Second column of U:** Emphasizes precipitation, especially in winter months, with positive values concentrated in western and northern coastal regions and negative values in eastern and southern areas. $V^T$ weights confirm precipitation's dominance with weight contributions of 0.28-0.31 from September to April and 0.11-0.20 in remaining months. This represents *maritime versus continental climate effects*.

**Third column of U:** Distinguishes areas with hot, wet summers and cold, dry winters (positive values) from coastal areas, especially western shores, with moderate climates (negative values). $V^T$ highlights precipitation in May-August (e.g., 0.41 for May and 0.46 for June) and summer maximum temperatures (range 0.16-0.22). This captures *seasonal temperature amplitude and summer precipitation patterns*.

**Fourth column of U:** Identifies regions with high temperature amplitudes within the same month, rainy springs, and dry summers, such as Southern Europe, Northern Scandinavia, and Arctic Ocean islands (positive values), while Central Europe shows more stable monthly temperatures (negative values). $V^T$ indicates strong variation in temperatures (weights from $-0.10$ to $-0.25$ for minimum temperatures and 0.10 to 0.22 for maximum temperatures) and precipitation (negative in summer, positive in spring).

This represents *continental effects and winter extremes.*

**Fifth column of U:** Represents regions with greater differences between summer and winter temperatures and precipitation, with positive values in the northeast and southeast and negative values in Western Europe, Iceland, and northern islands. $V^T$ highlights seasonal rainfall and temperature variations, with large positive values (0.18–0.23) for summer temperatures and winter rainfall and large negative values ($-0.14$ to $-0.24$) for the opposite pattern. This captures *seasonal precipitation reversal and microclimatic variations.*

### 2.4 Interpretation of Left Singular Vectors Based on North-South and East-West Distinction

Figures show the first column of $U$ plotted against the other left singular vectors (up to the fifth), with data points colored by their North-South and East-West locations respectively.

The first singular vector $U_1$ defines distinct clusters: Arctic Ocean islands, Northern Europe, and Southern Europe. The influence of other singular vectors (on the y-axis) slightly adjusts the clusters but doesn't disrupt the core grouping, reflecting the dominance of the first singular value ($s_1 = 290$ against $s_2 = 151$ and lower values). The East-West location does not significantly impact clustering in $U_1$, as points from both directions overlap.

$U_2$ captures more of the East-West distribution. When plotted against $U_3$, Eastern points tend to have lower $U_2$ values compared to Western ones, though the difference in $U_3$ is less pronounced: any data point can have low and moderate values, but only western and central points show large values. A small cluster of Eastern and Northern points shows both $U_2$ and $U_3$ values near their minimums, but overall, no strong clustering pattern is observed beyond the East-West distinction.

### 2.5 Rank Selection Methods

The suggested ranks for truncated SVD from various methods are summarized in Table 1, ranging from $k = 1$ (entropy-based method) to $k = 37$ (Guttman–Kaiser criterion).

| Method | Selected Rank |
|---|---|
| Guttman–Kaiser criterion | 37 |
| 90% of squared Frobenius norm | 3 |
| Scree test | 6 |
| Entropy-based method | 1 |
| Random flipping of signs | 7 |

Table 1: Results of different rank selection methods for truncated SVD

For the **Scree test**, we selected $k = 6$ as the elbow point, where singular values begin to flatten: although the change between $k = 4$ and $k = 5$ is minimal, the drop between

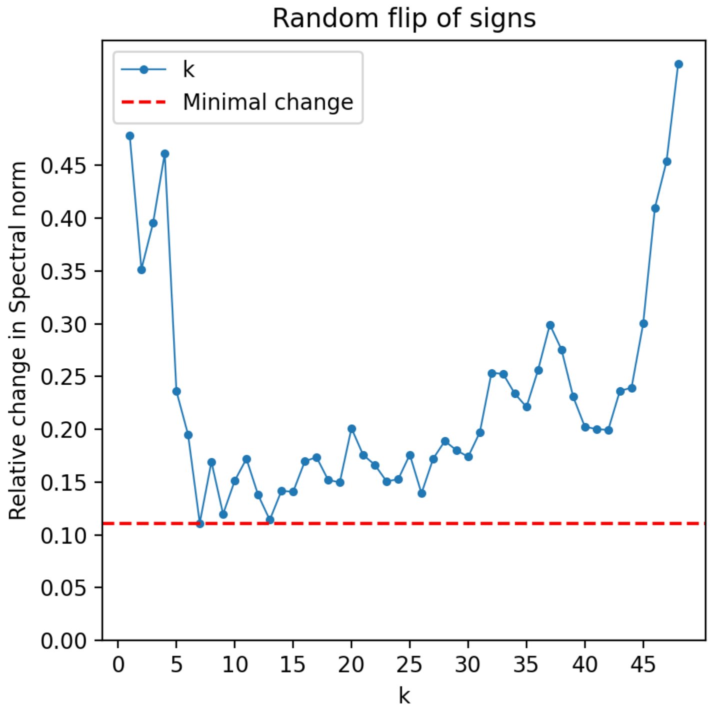


Figure 2: Random flip of signs method for selection of k

$k = 5$ and $k = 6$ remains noticeable. This choice aligns with the result of a knee locator algorithm, which identifies the maximum curvature at $k = 6$.

For the **random flipping of signs**, the minimum relative change (0.111) in spectral norms of the original and randomly perturbed residual matrices occurs at $k = 7$, suggesting this rank as optimal.

Given the variability across methods, we avoid blindly relying on any single suggestion. Based on the $\Sigma$ matrix analysis and previous examination of the first five singular vectors in Sections 2.3 and 2.4, we recommend exploring $k$ between **5 and 6**. This range balances capturing essential climate structure while avoiding noise, ensuring a compact yet informative representation suitable for downstream analysis.

## 2.6 SVD and Noise

Figure 4 shows the impact of rank $k$ and noise level $\varepsilon$ on RMSE for rank-$k$ truncated SVD reconstruction. For $k = 1$, RMSE is highest at low noise levels ($\approx 0.56$) and remains nearly constant as noise increases. This occurs because the first singular vector captures only dominant data features; although insufficient to capture all subtleties, it remains unaffected by noise.

For higher $k$, RMSE decreases at low noise levels, reflecting improved reconstruction

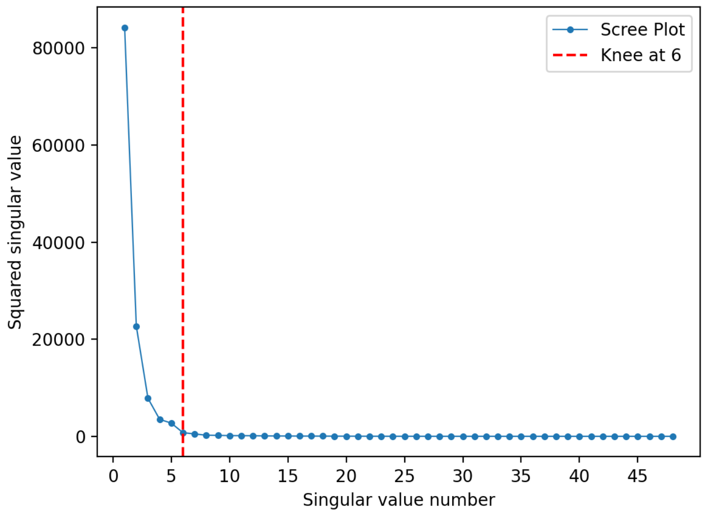


Figure 3: Cattell's Scree test with KneeLocator algorithm

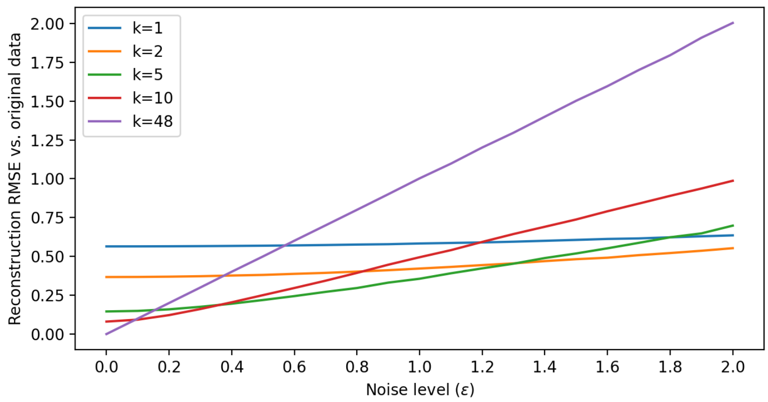


Figure 4: Impact of Noise on RMSE for Rank-k Approximations

accuracy. However, as noise level increases, RMSE grows faster for larger $k$, as more singular vectors begin to incorporate noise. Specifically:

- $k = 2$ shows RMSE $\approx 0.36$ at $\varepsilon = 0$, increasing gradually to $\approx 0.56$ at $\varepsilon = 2$.
- $k = 5$ demonstrates RMSE from $\approx 0.15$ to $\approx 0.68$, with moderate sensitivity.
- $k = 10$ exhibits stronger noise sensitivity, with RMSE from $\approx 0.08$ to $\approx 0.98$.
- Full-rank reconstruction ($k = 48$) perfectly approximates clean data (RMSE $\approx 0$) but is most sensitive to noise, reaching RMSE $\approx 2.0$ at high noise levels.

Notably, at $\varepsilon > 1.4$, even $k = 1$ outperforms $k = 48$, demonstrating the catastrophic failure of full-rank reconstruction under high noise conditions.

In practical settings, the noise level is often unknown. We recommend finding a tradeoff depending on expected noise level: smaller $k$ (e.g., $k = 2$ for this case) minimizes noise sensitivity but sacrifices detail, while moderate $k$ (e.g., $k = 5$ or $k = 10$) balances reconstruction quality and robustness to noise. This illustrates the fundamental bias-variance tradeoff in dimensionality reduction.

# 3 SVD and Clustering

## 3.1 Plotting Clusters against their Coordinates

As a first step, the normalized data $X$ was clustered using the K-Means Algorithm with $k = 5$. Then, Figure 5 was created with the x-coordinates of the places on the x-axis and the y-coordinates of the places on the y-axis. The colour corresponds to the cluster.

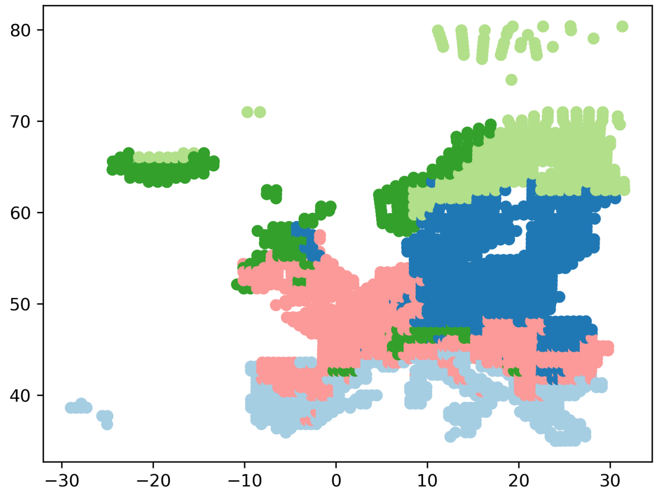


Figure 5: Clusters by Coordinates

Even though the entire (normalized) dataset has been used for clustering, two components can explain already important aspects of the clustering. Points next to each other more often than not belong to the same cluster. There is a clear separation of the clusters depending on the latitude (the y-coordinate) of the place. This makes sense, as the latitude is crucial for the climate, i.e. temperature and rainfall. However, some clusters are rather scattered, as the dark green one. This is contrary to the procedure of K-Means, which produces centroid based clusters in the end and a corresponding Voronoi-Diagram. Hence, these clusters cannot be fully explained by using only the information from longitude and latitude.

### 3.2 Plotting Clusters against their Principal Components

In the second part of the task, I created a similar plot, but used the first two left singular vectors as x- and y- coordinate of the plot.

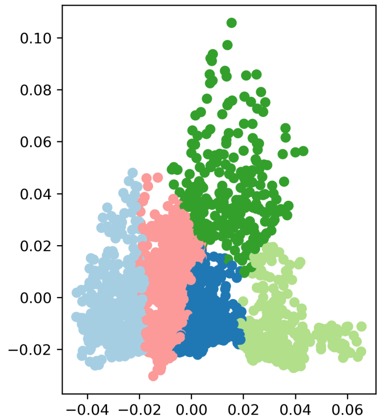


Figure 6: Clusters by first two Singular Vectors

This plot explains the clusters much better than the previous one using only the longitude and latitude. It clearly depicts the Voronoi-Diagram typical to K-Means Clustering. This is the expected result, as the first two singular vectors should be the two dimensions that can explain most of the variance of the dataset. However, the price to pay is that the coordinatees of the diagram are not intutive anymore. Compared to the previous figure with longitude and latitude, the first and second singular vectors are linear combinations of multiple features, leading to an abstract quantity that cannot be interpreted. The clusterss all seem to be close to each other, and there are no clear outliers.

### 3.3 Clustering after applying SVD

In the last part of this task, the dimension of the dataset was first reduced using PCA, then the plotting of task a) was repeated. The corresponding plot can be found in the appendix.

One can see that the clustering using the size-1 (10) reduction of the dataset results in similar clusters as the clustering using the entire dataset, not to speak of the size-2 (11) and size-3 (12) reduction. There are only minor differenes identifiable, especially in the region of England (the left upper corner of the mainland is coloured dark green in the original data, but partly light blue in the rank-2 and light green in the size-2 approximation). Another minor modification is in the middle of the light blue cluster on the right of the mainland. In the rank-1 and rank-2 approximation, it contains some green and red stains, whereas in the original data it is solely light blue. Besides these two little details, the plots seem the same. Especially, there is no difference identifiable between the clustering using the rank-3 approxmation and the clustering using the original data.

# 4 Appendix

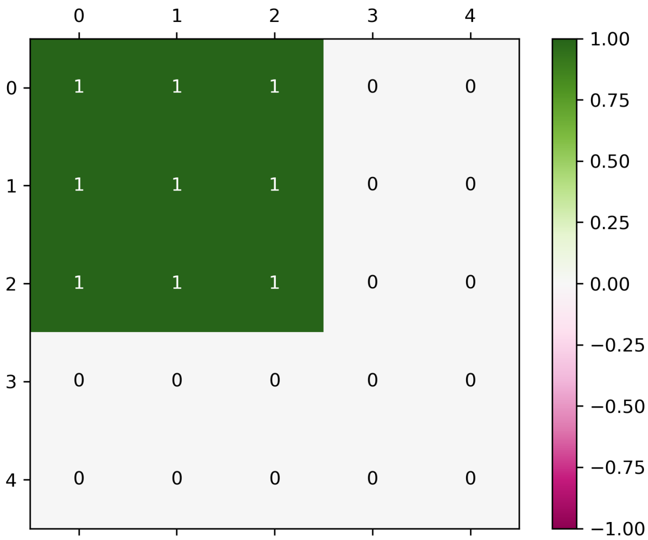


Figure 7: Rank 1 approximation of Matrix 4

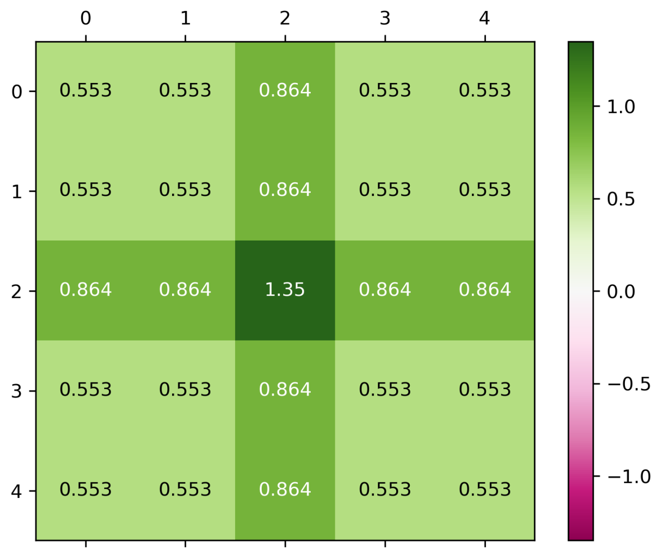


Figure 8: Rank 1 approximation of Matrix 5

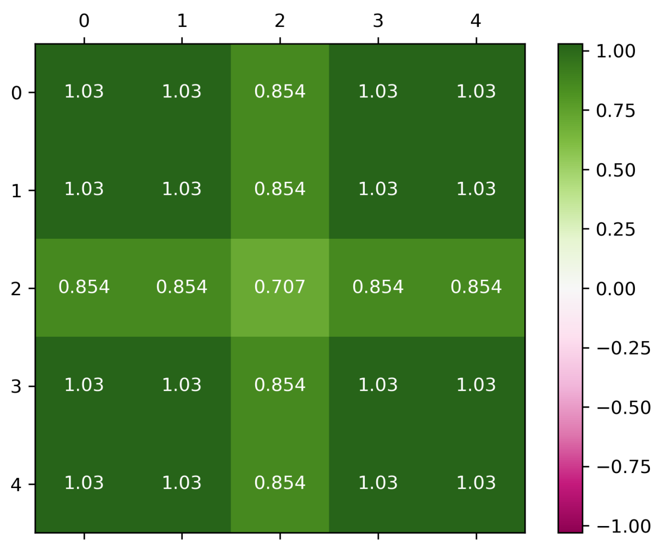


Figure 9: Rank 1 approximation of Matrix 6

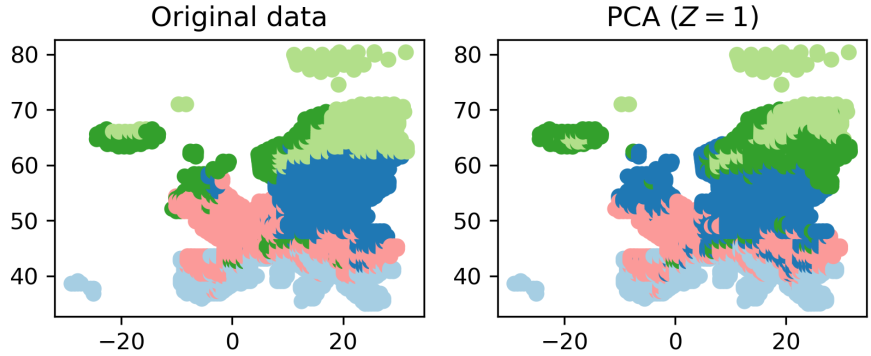


Figure 10: Rank-1 approximation

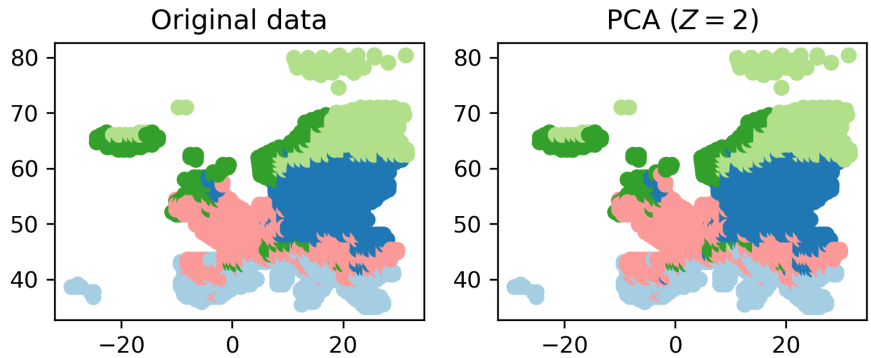


Figure 11: Rank-2 approximation

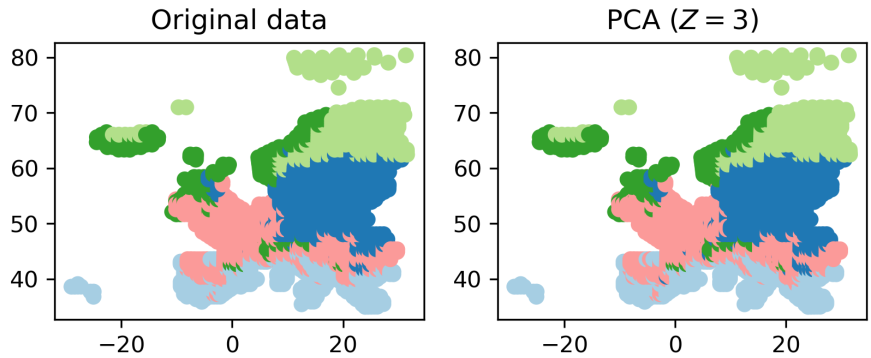


Figure 12: Rank-3 approximation

Figure 13: Clustering: Comparing Original Dataset with SVD Approximation

## Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Ich bin mir bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

**Declaration of Used AI Tools**

| Task | Solved with AI? | Tool | Purpose | Useful? |
|---|---|---|---|---|
| 1 | Yes | Gemini | Create LaTeX Expressions | yes |
| 2c | Yes | ChatGPT | Generate draft solution | yes |
| 2e | Yes | ChatGPT | Generate draft solution | yes |
| 3 | Yes | Gemini | Create LaTeX Expressions | yes |

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